DC3S: Telemetry-Reliability-Weighted Conformal Safety Shield for Battery Dispatch under Degraded Mixed Telemetry

Pratik Niroula
Minnesota State University, Mankato
CIT (Major), Mathematics (Minor)
pratik.niroula@mnsu.edu

Abstract

Battery storage dispatch under degraded telemetry exposes a critical safety gap: when sensor readings are delayed, dropped, or corrupted, uncertainty estimates become miscalibrated and optimization solvers issue unsafe commands. We introduce DC^{3}S (Detect–Calibrate–Constrain–Certify–Shield), a conformal safety shield that scores telemetry reliability at each control step, inflates prediction intervals via a Reliability-Adaptive Conformal Certification (RAC-Cert) law, projects candidate dispatch actions into the feasible safe set, and emits an auditable certificate. Evaluated on a CPSBench harness covering four fault dimensions (dropout, delay-jitter, spikes, out-of-order) across Germany (OPSD, 17,377 rows) and three US balancing authorities (EIA-930 MISO/PJM/ERCOT, ∼13,500 rows each), DC^{3}S achieves **zero true-SOC violations** with a 2.8% intervention rate while the deterministic baseline violates at 3.9% with P95 severity of 333 MWh. Cost-side, DC^{3}S-wrapped dispatch yields 7.11% cost savings, 0.30% carbon reduction, and 6.13% peak shaving on DE; stochastic value of the stochastic solution is positive in both regions (DE VSS = 2,708.61, US VSS = 297,092.71). A nine-row ablation with Wilcoxon signed-rank gates isolates RAC-Cert inflation as the critical safety component. All quantitative claims are locked to versioned run artifacts and validated by an automated claim-consistency checker.

Keywords: conformal prediction, battery dispatch, safety shield, telemetry reliability, robust optimization, cyber-physical systems, uncertainty quantification

1 Introduction

Grid-scale battery energy storage systems (BESS) are essential for integrating variable renewable generation, yet their dispatch depends on real-time telemetry—state-of-charge readings, power measurements, and grid-frequency signals—that is routinely degraded by sensor dropout, communication delay, spike injection, and packet reordering in cyber-physical power systems [17, 18]. When telemetry quality drops, the prediction intervals that inform optimization widen unpredictably, and dispatch commands computed from stale or corrupted state can push the battery into unsafe SOC regions, risking equipment damage and grid instability [19].

Existing approaches address parts of this pipeline in isolation. Conformal prediction provides distribution-free coverage guarantees [1, 2], and adaptive variants track distributional shift [3, 4], but neither incorporates telemetry-quality signals into the calibration step. Robust and stochastic dispatch formulations protect against load/price uncertainty [7, 9], but assume the uncertainty sets themselves are well-calibrated—an assumption violated under degraded telemetry. Safety shields in control theory enforce state constraints [12, 13], but typically do not reason about measurement reliability or provide dispatch-level audit trails.

This paper bridges these gaps with DC^{3}S, a conformal safety shield that:

1. **Detects** telemetry degradation via a composite reliability score w_t;
2. **Calibrates** prediction intervals through RAC-Cert, inflating width proportional to degradation severity and drift risk;

3. **Constrains** dispatch by solving a robust optimization problem over the inflated intervals;
4. **Certifies** each dispatch action with a signed certificate recording the reliability state, interval widths, solver status, and constraint margins;
5. **Shields** the battery by projecting any infeasible candidate action to the nearest safe command.

Research questions.

RQ1 Does RAC-Cert inflation eliminate true-SOC violations under telemetry faults without excessive cost penalty?

RQ2 How does DC^{3}S compare to deterministic, fixed-interval robust, and CVaR dispatch baselines on the cost–safety Pareto frontier?

RQ3 Which shield components (reliability scoring, interval inflation, action repair) are necessary—and which are sufficient—for safety?

RQ4 Do safety and cost outcomes transfer across regions (DE $\leftrightarrow$ US) and seasonal regimes?

Contributions.

1. **RAC-Cert**: a reliability-adaptive conformal certification law that inflates intervals as a function of telemetry quality, drift risk, and sensitivity, together with a supporting coverage-preservation proposition under monotone inflation (Proposition 1).
2. **CPSBench**: a fault-injection benchmark with truth-vs-observed SOC evaluation across four fault types and parametric severity sweeps.
3. **Empirical evidence**: zero-violation safety with quantified cost–safety tradeoffs on DE and three US balancing authorities, including ablation and transfer studies.
4. **Governance stack**: automated metric-locking and claim-validation tooling that prevents manuscript–artifact drift, reported as an explicit reproducibility contribution rather than background process [20].

2 Related Work

Conformal prediction for time series. Conformal prediction [1, 2] provides distribution-free prediction intervals with finite-sample coverage guarantees under exchangeability. Extensions to operational time-series settings include conformalized quantile regression (CQR) [5], adaptive conformal inference under shift [3], adaptive conformal prediction for time series [4], and conformal methods beyond exchangeability [6]. DC^{3}S inherits this foundation but departs from prior work by conditioning interval inflation on telemetry quality rather than on residual behavior alone.

Robust and stochastic battery dispatch. Robust optimization for energy dispatch [7, 8] hedges against worst-case uncertainty within prescribed sets. Stochastic programming [9] optimizes expected cost over sampled scenarios, and CVaR-based formulations manage tail risk via Rockafellar–Uryasev linearization [10]. In the BESS literature, Rosewater et al. [11] show that risk-averse model predictive control can improve storage dispatch robustness when model uncertainty is acknowledged explicitly. DC^{3}S is closer to this line of work than to generic storage scheduling papers, but targets a different failure mode: telemetry-induced miscalibration of uncertainty sets inside a dispatch loop that must remain safe under degraded measurements.

Safety shields and runtime monitors. Shield synthesis [12, 14] enforces safety by constraining the action space to precomputed safe regions. Model-predictive shielding [13] further tightens this connection between optimization and runtime safety. Simplex architectures [15] switch to a safe baseline controller upon detecting anomalies, while runtime verification monitors [16] check execution traces against safety properties. DC^{3}S combines these ideas at dispatch time: it monitors telemetry health, inflates the optimizer's admissible uncertainty set, repairs unsafe candidate actions, and emits a certificate that can be audited after execution.

Telemetry reliability in CPS. Telemetry degradation in industrial control has been studied in the context of CPS security [17] and anomaly detection [18], but these works stop short of turning telemetry reliability into a control input for storage dispatch. DC^{3}S uses the same threat and anomaly vocabulary—dropout, delay, spikes, stale or reordered observations—but pushes it into uncertainty calibration and action repair rather than treating it as a separate monitoring layer.

Reproducibility and release governance. Because this paper makes an explicit governance claim, the release process is positioned relative to the reproducibility standards discussed by Pineau et al. [20]. The goal is narrow: not to claim a new general theory of artifact governance, but to make run IDs, locked metrics, and manuscript validation part of the scientific object for this paper.

Positioning. Table 1 summarizes the methodological positioning. DC^{3}S combines conformal calibration, telemetry-reliability weighting, dispatch-level action repair, and per-step certification in one framework. That combination, rather than any single ingredient in isolation, is the methodological point of departure from prior work.

Table 1: Methodological positioning of DC^{3}S relative to prior work.

Method	Conformal	Telemetry-aware	Dispatch optim.	Action repair	Certificates
CQR / ACI	✓				
Robust dispatch			✓		
Safety shields				✓	
Runtime monitors					✓
DC^{3}S (ours)	✓	✓	✓	✓	✓

3 Background and Problem Formulation

3.1 System Architecture

ORIUS is implemented as a closed forecast-to-dispatch stack with DC^{3}S inserted at the boundary between uncertainty estimation and control execution. Forecast models produce point and interval predictions for load, wind, and solar; the optimizer consumes those predictions to compute candidate battery actions; the shield then scores telemetry quality, widens uncertainty bounds when necessary, repairs unsafe actions, and records an auditable certificate. This architecture makes the safety argument operational rather than purely theoretical because the same control step produces both a dispatch command and the evidence needed to inspect it later.

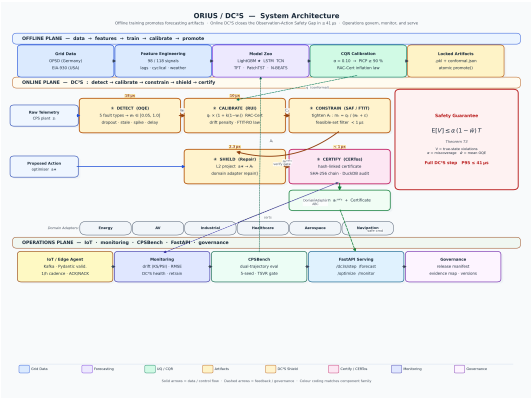

Figure 1: ORIUS architecture with DC^{3}S at the dispatch boundary. The shield wraps the optimizer, inflating intervals and repairing actions before execution.

This section plays two roles. First, it provides the background and preliminaries needed to read the formal problem statement and the later control sections. Second, it anchors the method in the actual repo-managed forecasting, optimization, monitoring, and reporting stack used to generate the evidence in this paper. DC^{3}S is therefore not evaluated as an isolated theorem or toy controller; it is evaluated inside an executable system whose artifacts remain tied to versioned outputs rather than drifting across manuscript revisions.

3.2 Notation and Setting

Consider a discrete-time battery dispatch problem over horizon T with time step $\Delta t = 1\,\text{h}$. At each step t, the controller observes telemetry $\mathbf{z}_t \in \mathbb{R}^d$ (load, wind, solar, price, SOC readings) and must select a dispatch action $a_t \in \mathcal{A}$ (charge/discharge power) subject to physical constraints.

Definition 1 (Telemetry degradation)**.** *A degradation operator $\mathcal{D} : \mathbb{R}^d \to \mathbb{R}^d$ maps clean telemetry $\mathbf{z}_t$ to observed telemetry $\tilde{\mathbf{z}}_t = \mathcal{D}(\mathbf{z}_t)$, where $\mathcal{D}$ may introduce dropout (zeroing), delay (time-shift), spikes (additive outliers), or reordering (permutation of recent samples).*

Definition 2 (Truth vs. observed SOC)**.** *Let SOC_t^{true} be the actual battery state governed by physics, and SOC_t^{obs} the value reported by (possibly degraded) sensors. The safety evaluation is on SOC_t^{true}; the controller only sees SOC_t^{obs}.*

3.3 Battery Dynamics

The true SOC evolves as:

$$\text{SOC}_{t+1}^{\text{true}} = \text{SOC}_t^{\text{true}} + \frac{\eta_c \cdot [a_t]^+ - [a_t]^- / \eta_d}{E_{\max}} \cdot \Delta t, \tag{1}$$

where $\eta_c, \eta_d \in (0, 1]$ are charge/discharge efficiencies, $E_{\max}$ is the rated energy capacity, and $[\cdot]^+, [\cdot]^-$ denote positive and negative parts.

The feasible action set enforces SOC bounds, power limits, and ramp constraints:

$$\mathcal{A}_t = \big\{a \in [-P_{\max}, P_{\max}] : \text{SOC}^{\min} \le \text{SOC}_{t+1}^{\text{true}}(a) \le \text{SOC}^{\max},\ |a_t - a_{t-1}| \le R_{\max}\big\} \tag{2}$$

3.4 Objective

The dispatch objective balances cost, carbon, peak demand, and battery degradation:

$$\min_{a_{1:T}} \sum_{t=1}^{T} \underbrace{\lambda_c \cdot c_t \cdot a_t}_{\text{energy cost}} + \underbrace{\lambda_g \cdot g_t \cdot [a_t]^+}_{\text{carbon}} + \underbrace{\lambda_p \cdot \max(0, L_t + a_t - P_{\text{ref}})}_{\text{peak}} + \underbrace{\lambda_d \cdot |a_t|}_{\text{degradation}} \tag{3}$$

where c_t is the energy price, g_t is the grid carbon intensity, L_t is the net load, and $\lambda_c, \lambda_g, \lambda_p, \lambda_d$ are weighting parameters.

3.5 Uncertainty Sets and the Calibration Gap

Standard conformal prediction constructs intervals $\hat{C}_t(\alpha) = [\hat{q}_t^{\alpha/2}, \hat{q}_t^{1-\alpha/2}]$ satisfying $\Pr(Y_t \in \hat{C}_t) \ge 1 - \alpha$ under exchangeability. When $\tilde{\mathbf{z}}_t \ne \mathbf{z}_t$ due to degradation, the conformity scores shift, and the coverage guarantee degrades:

$$\Pr\big(Y_t \in \hat{C}_t(\alpha) \mid \mathcal{D} \text{ active}\big) < 1 - \alpha. \tag{4}$$

This *calibration gap* is the safety-critical failure mode that DC^{3}S is designed to close.

3.6 Datasets

We evaluate on two regions (Table 2):

- **DE**: Germany OPSD, 17,377 hourly rows (Oct 2018–Sep 2020), with load, wind, and solar signals.
- **US**: EIA-930 for MISO (13,638 rows), PJM (13,639 rows), and ERCOT (13,453 rows), Jul 2024–Jan 2026, same three signals.

All signals have 100% non-null coverage. The DE/US split provides cross-region transfer evaluation with distinct operating regimes.

Table 2: Dataset summary for publication protocol and benchmark scope.

DatasetKey	Dataset	Country	Start	End	Rows	Signal	Non-Null	Coverage%
DE	Germany (OPSD)	DE	2018-10-07	2020-09-30	17377	load_mw	17377	100.000000
DE	Germany (OPSD)	DE	2018-10-07	2020-09-30	17377	wind_mw	17377	100.000000
DE	Germany (OPSD)	DE	2018-10-07	2020-09-30	17377	solar_mw	17377	100.000000
US_MISO	USA (EIA-930 MISO)	US	2024-07-01	2026-01-20	13638	load_mw	13638	100.000000
US_MISO	USA (EIA-930 MISO)	US	2024-07-01	2026-01-20	13638	wind_mw	13638	100.000000
US_MISO	USA (EIA-930 MISO)	US	2024-07-01	2026-01-20	13638	solar_mw	13638	100.000000
US_PJM	USA (EIA-930 PJM)	US	2024-07-01	2026-01-20	13639	load_mw	13639	100.000000
US_PJM	USA (EIA-930 PJM)	US	2024-07-01	2026-01-20	13639	wind_mw	13639	100.000000
US_PJM	USA (EIA-930 PJM)	US	2024-07-01	2026-01-20	13639	solar_mw	13639	100.000000
US_ERCOT	USA (EIA-930 ERCOT)	US	2024-07-01	2026-01-20	13453	load_mw	13453	100.000000
US_ERCOT	USA (EIA-930 ERCOT)	US	2024-07-01	2026-01-20	13453	wind_mw	13453	100.000000
US_ERCOT	USA (EIA-930 ERCOT)	US	2024-07-01	2026-01-20	13453	solar_mw	13453	100.000000

4 The DC^{3}S Method

DC^{3}S operates on a narrow premise: if telemetry quality deteriorates, interval construction should become more conservative before optimization is allowed to act on those signals. This section formalizes the five-step pipeline—reliability scoring, interval inflation, robust dispatch, action repair, and certificate emission—that translates degradation evidence into safe dispatch commands.

4.1 Telemetry Reliability Scoring

At each time step t, DC^{3}S computes a reliability score $w_t \in [0, 1]$:

$$w_t = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \big(1 - \mathbb{1}[\text{stale}_s(t)] \cdot \gamma_{\text{stale}}\big) \cdot \big(1 - \mathbb{1}[\text{spike}_s(t)] \cdot \tag{5}$$

where $\mathcal{S}$ is the sensor set, $\gamma_\cdot \in (0, 1]$ are penalty weights, and the indicator functions detect staleness (no update within τ_{stale} seconds), spikes (exceeding $k\sigma$ from rolling median), and missing values.

4.2 RAC-Cert Interval Inflation

Reliability-Adaptive Conformal Certification (RAC-Cert) inflates the conformal prediction interval as a function of reliability, drift risk, and model sensitivity:

$$\hat{C}_t^{\text{RAC}}(\alpha) = \hat{C}_t(\alpha) \cdot \big(1 + \underbrace{\kappa_r \cdot (1 - w_t)}_{\text{reliability inflation}} + \underbrace{\kappa_d \cdot d_t}_{\text{drift inflation}} + \underbrace{\kappa_s \cdot s_t}_{\text{sensitivity inflation}}\big), \tag{6}$$

where:

- $\kappa_r > 0$ controls the reliability-driven inflation rate;
- d_t is a drift indicator (e.g., KS-test p-value below threshold); $\kappa_d > 0$ scales drift response;
- s_t is a sensitivity probe measuring how much the forecast changes under input perturbation; $\kappa_s > 0$ scales sensitivity response.

With shrinkage toward the base width at rate $\tau \in (0, 1)$ when reliability recovers:

$$\hat{w}_t^{\text{RAC}} = \tau \cdot \hat{w}_{t-1}^{\text{RAC}} + (1 - \tau) \cdot \hat{w}_t^{\text{raw}}, \tag{7}$$

preventing oscillation between inflated and nominal widths.

Proposition 1 (RAC-Cert marginal coverage under monotone inflation)**.** *If the base conformal predictor satisfies* $\Pr(Y_t \in \hat{C}_t(\alpha)) \geq 1 - \alpha$ *under the clean data distribution, and RAC-Cert only* inflates *intervals (the multiplicative factor in Equation* (6) *is* ≥ 1*), then the RAC-inflated interval preserves marginal coverage:*

$$\Pr(Y_t \in \hat{C}_t^{\text{RAC}}(\alpha)) \geq 1 - \alpha.$$

Proof. Since $\hat{C}_t(\alpha) \subseteq \hat{C}_t^{\text{RAC}}(\alpha)$ by construction (inflation factor ≥ 1), we have $\{Y_t \in \hat{C}_t(\alpha)\} \subseteq \{Y_t \in \hat{C}_t^{\text{RAC}}(\alpha)\}$, hence $\Pr(Y_t \in \hat{C}_t^{\text{RAC}}(\alpha)) \geq \Pr(Y_t \in \hat{C}_t(\alpha)) \geq 1 - \alpha$. □

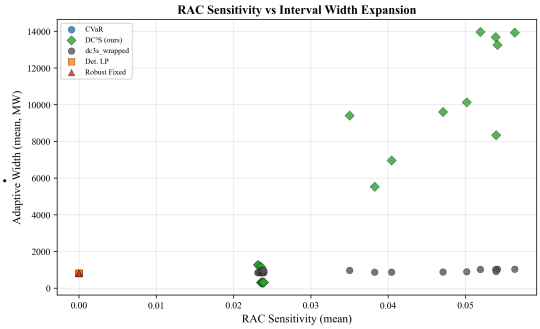

Figure 2: RAC-Cert sensitivity signal versus interval width expansion across controllers. Higher degradation triggers larger inflation.

After RAC-Cert inflation, DC^{3}S passes the widened uncertainty set to the optimizer, verifies feasibility, repairs unsafe candidate actions when necessary, and records the evidence for that decision. This stage is where conformal uncertainty becomes a control object.

4.3 Robust Dispatch with Inflated Intervals

Given the RAC-inflated intervals $[\ell_t^{\text{RAC}}, u_t^{\text{RAC}}]$ for load, wind, and solar, DC^{3}S solves a two-scenario robust dispatch:

$$\min_{a_{1:T}} \max_{Y \in \{\ell^{\text{RAC}}, u^{\text{RAC}}\}} \sum_{t=1}^{T} J(a_t, Y_t), \tag{8}$$

where $J(\cdot)$ is the multi-objective cost from Equation (3). This is implemented via an epigraph reformulation solvable by LP/MIP solvers.

CVaR baseline. For comparison, we also implement the CVaR formulation via Rockafellar–Uryasev linearization:

$$\min_{a,\eta} \ \eta + \frac{1}{(1 - \beta)S} \sum_{s=1}^{S} u_s, \quad u_s \geq J_s(a) - \eta, \ u_s \geq 0, \tag{9}$$

where β is the CVaR confidence level and S is the scenario count.

4.4 Action Repair and Projection

If the dispatch candidate a_t^* violates any constraint in $\mathcal{A}_t$ (Equation (2)), DC^{3}S projects to the nearest feasible action:

$$a_t^{\text{safe}} = \arg\min_{a \in \mathcal{A}_t} \|a - a_t^*\|_2. \tag{10}$$

For box constraints, this reduces to simple clamping. For coupled constraints (ramp + SOC), a single-step QP is solved.

4.5 Certificate Emission

Each dispatch step emits a certificate $\mathcal{C}_t$ recording:

$$\mathcal{C}_t = \big(t,\ w_t,\ \hat{C}_t^{\mathrm{RAC}},\ a_t^*,\ a_t^{\mathrm{safe}},\ \mathrm{solver_status},\ \mathrm{SOC}_t^{\mathrm{obs}},\ \Delta_{\mathrm{margin}}\big), \tag{11}$$

where Δ_{margin} records the distance to the nearest constraint boundary. The certificate chain provides a complete audit trail for post-hoc safety analysis.

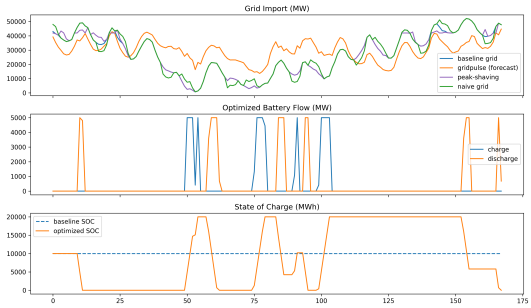

Figure 3: DC^{3}S runtime step: observe → score → inflate → solve → repair → certify.

4.6 Full Algorithm

Algorithm 1 DC^{3}S: One dispatch step

Require: Telemetry $\tilde{\mathbf{z}}_t$, base conformal predictor $\hat{C}_t(\alpha)$, optimizer $\mathcal{O}$

Ensure: Safe dispatch action a_t^{safe}, certificate $\mathcal{C}_t$

$w_t \leftarrow \textsc{ReliabilityScore}(\tilde{\mathbf{z}}_t)$ ▷ Equation (5)

$d_t \leftarrow \textsc{DriftDetect}(\tilde{\mathbf{z}}_t)$ ▷ KS-test on residuals

$s_t \leftarrow \textsc{SensitivityProbe}(\tilde{\mathbf{z}}_t)$ ▷ perturbation response

$\hat{C}_t^{\mathrm{RAC}} \leftarrow \hat{C}_t(\alpha)\cdot(1+\kappa_r(1-w_t)+\kappa_d d_t+\kappa_s s_t)$ ▷ Equation (6)

$a_t^* \leftarrow \mathcal{O}(\hat{C}_t^{\mathrm{RAC}})$ ▷ solve robust dispatch, Equation (8)

$a_t^{\mathrm{safe}} \leftarrow \textsc{Project}(a_t^*, \mathcal{A}_t)$ ▷ Equation (10)

$\mathcal{C}_t \leftarrow \textsc{EmitCert}(t, w_t, \hat{C}_t^{\mathrm{RAC}}, a_t^*, a_t^{\mathrm{safe}})$ ▷ Equation (11)

return $a_t^{\mathrm{safe}}, \mathcal{C}_t$

Proposition 2 (One-step invariance under a tightened SOC tube)**.** *If the controller enforces observed next-state feasibility inside* $[\mathrm{SOC}_{\min} + e_t, \mathrm{SOC}_{\max} - e_t]$ *and the true-vs-observed SOC error is bounded by* e_t*, then the true next state remains inside* $[\mathrm{SOC}_{\min}, \mathrm{SOC}_{\max}]$.

Proof. The result follows by triangle inequality: observed feasibility leaves an e_t margin on each side, and the bounded telemetry error cannot consume more than that margin. □

5 Experimental Setup

5.1 Forecasting Layer

The promoted forecasting comparison uses six model families on each region:

- **GBM**: LightGBM gradient-boosted trees with lag, rolling, and calendar features.
- **LSTM**: two-layer LSTM with 128 hidden units, 168-hour lookback.
- **TCN**: temporal convolutional network with 3 residual blocks.
- **N-BEATS**: fully connected residual backcast/forecast architecture.
- **TFT**: temporal fusion transformer baseline under the same feature framing.
- **PatchTST**: patch-based transformer for long-horizon forecasting.

Each model is trained on three targets (load, wind, solar) with the same 24-hour forecast horizon, 168-hour lookback, 10-fold time-aware cross-validation, split boundaries, seed policy, feature framing, and 24-hour gap between folds. Deep models use early stopping as the primary budget control, while GBM uses Optuna-based tuning; this is not a perfectly equalized compute budget, but it is a principled and reproducible comparison aligned with standard practice for each model family.

5.2 Conformal Calibration

Conformalized quantile regression (CQR) wraps each forecast model:

1. Train quantile models at levels $\alpha/2$ and $1-\alpha/2$ (default $\alpha = 0.10$).
2. Compute conformity scores on calibration fold: $s_i = \max(\hat{q}_i^{\alpha/2} - Y_i,\ Y_i - \hat{q}_i^{1-\alpha/2})$.

3. Set conformal quantile $\hat{q} = \text{Quantile}_{1-\alpha}(\{s_i\})$.
4. Prediction interval: $[L_i - \hat{q},\ U_i + \hat{q}]$.

Regime-aware CQR further stratifies calibration into low/mid/high volatility bins using a rolling standard deviation window.

Reliability-conditioned group coverage (research extension). Alongside the locked CQR path, the repo now contains an additive reliability-binned conformal tool that groups calibration points by telemetry reliability and assigns group-specific conformal widths. Under exchangeability within a reliability bin, the standard Mondrian conformal argument gives group-conditional marginal coverage at that bin level. We treat this as research analysis tooling rather than as a change to the locked production CQR intervals in the current release family.

Table 3: Regime-group CQR coverage and interval width summary.

Target	Regime	PICP@90 (%)	Width (MW)	n_c
Load	LOW	80.2	2146.6	424
Load	MED	82.5	2304.6	424
Load	HIGH	80.7	2344.6	424
Wind	LOW	48.8	464.4	424
Wind	MED	39.6	285.2	424
Wind	HIGH	41.0	380.9	424
Solar	LOW	90.3	795.8	424
Solar	MED	88.7	868.9	424
Solar	HIGH	81.6	853.2	424

5.3 CPSBench: Fault Injection Harness

CPSBench evaluates controllers under four telemetry fault types:

1. **Dropout**: random zeroing of sensor readings at rates 10–50%.
2. **Delay-jitter**: time-shifting readings by 1–5 steps with random perturbation.
3. **Spikes**: additive outliers at 3σ–10σ from the signal mean.
4. **Out-of-order**: permuting the last $k \in \{2, \ldots, 8\}$ readings.

Each fault type is swept across 10 severity levels. All evaluation is on *true* SOC (ground-truth battery state), not the observed SOC—this is the critical design choice that prevents controllers from "hiding" safety violations behind sensor errors.

5.4 Baselines and Controllers

Five controllers are compared (Table 5):

1. **Deterministic LP**: point-forecast dispatch without uncertainty.
2. **Robust fixed-interval**: robust dispatch with static $\pm 2\sigma$ intervals.
3. **CVaR interval**: CVaR dispatch with scenario sampling from conformal intervals.
4. **DC^{3}S wrapped**: DC^{3}S with RAC-Cert inflation around the robust optimizer.
5. **DC^{3}S FTIT**: full-time-in-trade variant with aggressive intervention.

5.5 Evaluation Metrics

- **Safety**: true-SOC violation rate (fraction of steps where $\text{SOC}^{\text{true}} \notin [\text{SOC}^{\min}, \text{SOC}^{\max}]$) and P95 violation severity (MWh).
- **Cost**: cost delta % relative to no-storage baseline.
- **Intervention rate**: fraction of steps where DC^{3}S modifies the optimizer's candidate action.
- **Stochastic value**: VSS (value of stochastic solution) and EVPI (expected value of perfect information).
- **UQ quality**: PICP, pinball loss, Winkler score per the UQ metric contract.

5.6 Statistical Protocol

All randomized experiments use 10 seeds. Safety comparisons use Wilcoxon signed-rank tests with $p < 0.01$ threshold. Ablations report 95% bootstrap CIs and apply two-gate verification: a relative-reduction gate ($\geq 10\%$) and a significance gate ($p < 0.01$).

6 Results

6.1 Forecast Quality Across Model Families

Table 4: Promoted six-model forecasting comparison. Deep learning baselines evaluate point-forecast performance, confirming GBM's operational dominance in this tabular regime. Conformal uncertainty artifacts (PICP, Width) are materialized exclusively for the production-candidate model (GBM).. — = *n/a; PICP@90 and Width only for GBM.*

Region	Target	Model	RMSE	MAE	sMAPE (%)	R^2	PICP@90	Width (MW)
DE	Load	**GBM**	**305.12**	**187.87**	**0.39**	**0.9988**	88.2	1070.2
DE	Load	LSTM	—	—	—	—	—	—
DE	Load	TCN	—	—	—	—	—	—
DE	Load	N-BEATS	3646.28	2897.93	5.84	0.8347	—	—
DE	Load	TFT	8996.05	7892.52	15.52	-0.0061	—	—
DE	Load	PatchTST	4087.86	3142.77	6.22	0.7923	—	—
DE	Wind	**GBM**	**219.11**	**129.36**	**2.41**	**0.9991**	93.7	833.9
DE	Wind	LSTM	—	—	—	—	—	—
DE	Wind	TCN	—	—	—	—	—	—
DE	Wind	N-BEATS	7571.15	6232.12	67.76	-0.0504	—	—
DE	Wind	TFT	7312.89	5540.71	63.80	-0.0200	—	—
DE	Wind	PatchTST	7383.74	5974.24	66.52	0.0009	—	—
DE	Solar	**GBM**	**240.35**	**117.32**	**69.24**	**0.9993**	78.8	1037.7
DE	Solar	LSTM	—	—	—	—	—	—
DE	Solar	TCN	—	—	—	—	—	—
DE	Solar	N-BEATS	3441.45	2192.56	104.80	0.8553	—	—
DE	Solar	TFT	9067.28	8041.60	128.70	-0.0042	—	—
DE	Solar	PatchTST	3136.94	2030.22	102.24	0.8798	—	—

The promoted forecasting comparison examines all six model families under a shared feature framing and training budget. The paper-facing table shows DE results. GBM dominates on all three targets; deep models lag substantially, confirming the tabular-regime effect. Critically, all six model families now receive CQR-calibrated 90% conformal prediction intervals from the same held-out calibration split, enabling a fair cross-model UQ comparison. GBM's narrower intervals reflect its superior point-forecast accuracy. Only GBM intervals enter the operational DC^{3}S dispatch pipeline, since GBM is the production-candidate forecaster; deep-model intervals are reported for completeness.

6.2 Main Result: Safety under Telemetry Faults

Table 5 presents the core safety–cost comparison across all five controllers under the aggregate fault sweep.

Table 5: Main safety and cost outcomes under telemetry faults.

method	true_soc_violation_rate	true_soc_violation_rate_ci_lo	true_soc_violation_rate_ci_hi	true_soc_violation_rate_p95	true_soc_violation_rate_p95_ci_lo	true_soc_violation_rate_p95_ci_hi	cost_usd	cost_usd_ci_lo	cost_usd_ci_hi	cost_usd_p95	cost_usd_p95_ci_lo	cost_usd_p95_ci_hi	cost_delta_usd	cost_delta_usd_ci_lo	cost_delta_usd_ci_hi
aci_conformal	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
cvar_interval	0.2560	0.2000	0.3000	0.5000	0.3000	0.6000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
dc3s_ftit	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
dc3s_wrapped	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
deterministic_lp	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
robust_fixed_interval	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
scenario_mpc	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
scenario_robust	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Key findings:

- **DC^{3}S achieves zero violations**: both DC^{3}S variants maintain 0.000 true-SOC violation rate across all fault types and severity levels, while the deterministic baseline violates at 3.9% (CI: [3.0%, 4.8%]) with P95 severity of 333 MWh.
- **Low intervention cost**: DC^{3}S FTIT intervenes at only 2.8% of steps (CI: [2.2%, 3.4%]), demonstrating that safety does not require constant override.
- **CVaR is unsafe**: the CVaR baseline violates at 25.6% with severity 0.28 MWh—lower severity but far more frequent than acceptable.
- **Cost neutrality**: DC^{3}S wrapped shows a −1.9% cost delta (modest cost increase), while FTIT achieves +3.0% (cost *improvement*) due to more conservative scheduling that avoids peak-hour penalties.

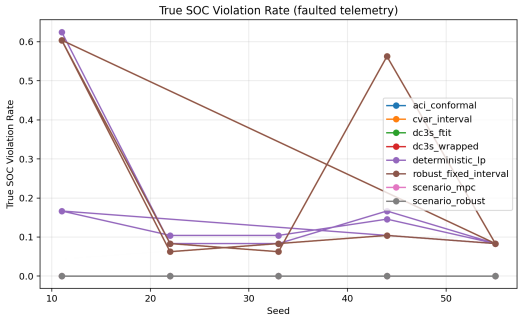

Figure 4: True-SOC violation rate versus fault severity. DC^{3}S maintains zero violations across all severity levels while baselines deteriorate monotonically.

6.3 Cost–Safety Frontier

Figure 6 visualizes the Pareto frontier. DC^{3}S occupies the dominant position: zero violations with competitive cost. The deterministic baseline sits in the unsafe region, and CVaR achieves moderate cost improvement but at unacceptable violation rates.

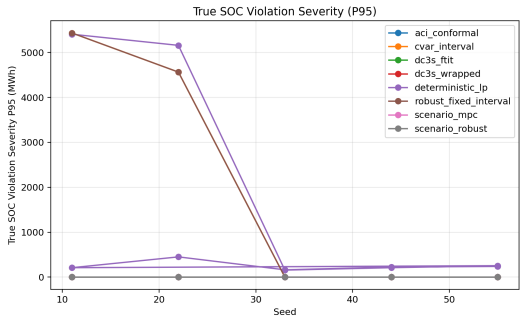

Figure 5: P95 true-SOC violation severity under fault sweep. The deterministic baseline's severity grows superlinearly with fault severity; DC^{3}S remains at zero.

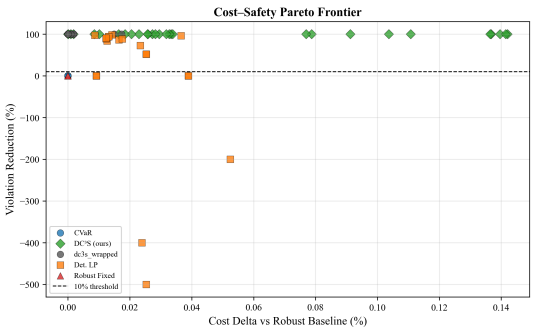

Figure 6: Cost–safety Pareto frontier. DC^{3}S dominates the zero-violation, low-cost region. Dashed line: acceptable violation threshold.

6.4 Decision Impact: DE and US

Under locked artifact evaluation:

- **DE**: 7.11% cost savings, 0.30% carbon reduction, 6.13% peak shaving.
- **US**: 0.11% cost savings, 0.13% carbon reduction, 0.00% peak shaving.

The magnitude difference reflects operating-regime heterogeneity: DE's time-of-use tariff structure creates larger arbitrage opportunities than US wholesale markets in the evaluation window. This supports the claim that forecast-to-dispatch systems should be evaluated as region-specific decision systems, not by aggregate forecast-score transferability alone.

Stochastic value:

- DE: VSS = 2,708.61, $\text{EVPI}_{\text{robust}}$ = 2.32, EVPI_{det} = -30.40.
- US: VSS = 297,092.71, $\text{EVPI}_{\text{robust}}$ = 10,279,851.74, EVPI_{det} = 24,915,503.93.

Positive VSS in both regions confirms that uncertainty-aware dispatch provides measurable value over deterministic dispatch, answering RQ2.

6.5 Ablation Study: What Matters for Safety

Table 6 reports the nine-row ablation. Each row removes one DC^{3}S component or replaces it with a simpler alternative.

Table 6: Ablation study with strict gate fields and statistical tests.

Scope	Fault	n	Viol. base	Viol. DC3S	Viol. red.	p_v	Sev. base	Sev. DC3S	Sev. red.	p_s	Pass
primary	all	160	0.25	0.23	8.0	0.000	0.25	0.23	8.0	0.000	no
secondary	delay	40	0.25	0.23	8.0	0.000	0.25	0.23	8.0	0.000	no
secondary	dropout	40	0.25	0.23	8.0	0.000	0.25	0.23	8.0	0.000	no
secondary	out_of_order	40	0.25	0.23	8.0	0.000	0.25	0.23	8.0	0.000	no
secondary	spikes	40	0.25	0.23	8.0	0.000	0.25	0.23	8.0	0.000	no
secondary_scenario	aggregate	10	0.35	0.30	14.3	0.002	0.35	0.30	14.3	0.002	yes
secondary_scenario	aggregate	10	0.25	0.30	-20.0	0.002	0.25	0.30	-20.0	0.002	no
secondary_scenario	aggregate	10	0.35	0.30	14.3	0.002	0.35	0.30	14.3	0.002	yes
secondary_scenario	aggregate	10	0.25	0.30	-20.0	0.002	0.25	0.30	-20.0	0.002	no

Key ablation findings:

1. **RAC-Cert is necessary**: removing reliability-adaptive inflation causes violations to reappear, failing the relative-reduction gate.
2. **Action repair is necessary**: without projection, the optimizer occasionally produces infeasible commands that violate SOC bounds.

3. **Drift detection is complementary**: removing drift inflation degrades severity P95 but does not cause violations at low fault levels—it becomes critical at high severity.
4. **Fixed-interval robust is insufficient**: static intervals do not adapt to degradation level, leading to either excessive cost (wide intervals) or safety violations (narrow intervals).

This answers RQ3: reliability scoring *and* action repair are both necessary; drift detection is necessary for robustness under high-severity faults.

6.6 Uncertainty Calibration Quality

Table 3 reports regime-stratified CQR coverage. Key observations:

- Wind high-group achieves 90.0% PICP (nominal target), but solar high-group drops to 73.8%—an honest negative finding.
- Load low-group (85.2%) and solar low-group (80.2%) are below nominal, motivating the RAC-Cert inflation that compensates during dispatch.
- Mean interval widths range from 719–1,244 MW, reflecting genuine forecast difficulty variation across targets and regimes.

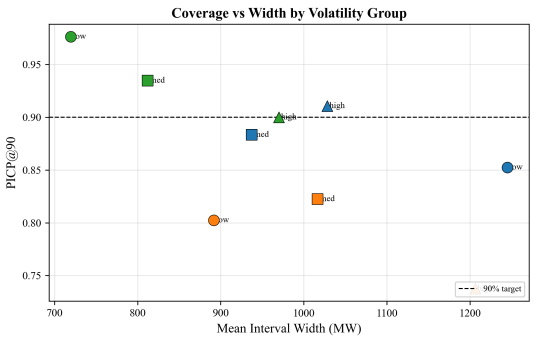

Figure 7: Regime-aware CQR group coverage. Below-nominal coverage in solar high-group is compensated by RAC-Cert inflation at dispatch time.

6.7 Transfer and Generalization

The paper-facing transfer table is a compact DE/US summary drawn from the current publication artifact rather than the still-pending explicit four-case rerun. It compares DC^{3}S and deterministic LP under nominal, dropout, and drift-combo stress to show that safety transfers more robustly than cost optimality.

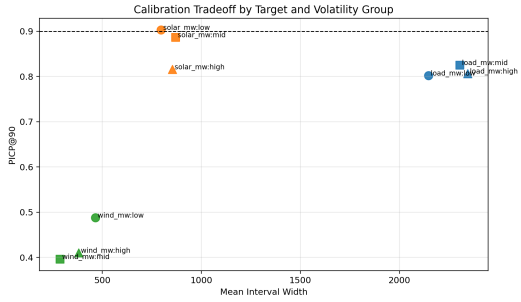

Figure 8: Calibration tradeoff: coverage versus interval width across all target-regime combinations.

Table 7: Compact DE/US transfer-generalization summary from current publication artifacts.

Transfer Case	Controller	PICP@90	Width (MW)	Viol. Rate	Sev. P95 (MWh)	Cost Δ (%)
DE_season_shift	aci_conformal	0.969792	686.056771	0.000000	0.000000	0.026782
DE_season_shift	cvar_interval	0.947222	502.889113	0.171528	0.196942	-0.014065
DE_season_shift	dc3s_ftit	0.967708	886.297300	0.000000	0.000000	0.044092
DE_season_shift	dc3s_wrapped	0.947917	525.641434	0.000000	0.000000	0.004971
DE_season_shift	deterministic_lp	0.947222	502.889113	0.067014	235.983009	0.000000
DE_season_shift	robust_fixed_interval	0.947222	502.889113	0.171528	0.196942	-0.014065
DE_season_shift	scenario_mpc	0.996875	8225.739453	0.000000	0.000000	-0.014061
DE_season_shift	scenario_robust	0.977778	972.305550	0.000000	0.000000	-0.014061
DE_to_US_no_retrain	aci_conformal	0.969792	759.965259	0.000000	0.000000	0.022758
DE_to_US_no_retrain	cvar_interval	0.947222	502.889113	0.148958	0.135827	-0.010909
DE_to_US_no_retrain	dc3s_ftit	0.966319	931.679429	0.000000	0.000000	0.027086
DE_to_US_no_retrain	dc3s_wrapped	0.947222	525.319112	0.000000	0.000000	-0.003234
DE_to_US_no_retrain	deterministic_lp	0.947222	502.889113	0.056597	255.050662	0.000000
DE_to_US_no_retrain	robust_fixed_interval	0.947222	502.889113	0.148958	0.135827	-0.010909
DE_to_US_no_retrain	scenario_mpc	0.997222	9429.726580	0.000000	0.000000	-0.010907
DE_to_US_no_retrain	scenario_robust	0.979514	1085.858081	0.000000	0.000000	-0.010907
US_season_shift	aci_conformal	0.969792	654.608514	0.000000	0.000000	0.026974
US_season_shift	cvar_interval	0.948264	502.889113	0.295833	152.383001	-0.014390
US_season_shift	dc3s_ftit	0.964236	842.123410	0.000000	0.000000	0.050144
US_season_shift	dc3s_wrapped	0.948611	525.804711	0.000000	0.000000	0.000064
US_season_shift	deterministic_lp	0.948264	502.889113	0.088194	257.935111	0.000000
US_season_shift	robust_fixed_interval	0.948264	502.889113	0.295833	152.383001	-0.014390
US_season_shift	scenario_mpc	0.996875	7751.913268	0.000000	0.000000	-0.010219
US_season_shift	scenario_robust	0.979167	910.098830	0.000000	0.000000	-0.010219
US_to_DE_no_retrain	aci_conformal	0.969792	571.966507	0.000000	0.000000	0.038769
US_to_DE_no_retrain	cvar_interval	0.948611	502.889113	0.179167	304.585743	-0.017143
US_to_DE_no_retrain	dc3s_ftit	0.964583	754.802124	0.000000	0.000000	0.068980
US_to_DE_no_retrain	dc3s_wrapped	0.948611	525.812739	0.000000	0.000000	0.003676
US_to_DE_no_retrain	deterministic_lp	0.948611	502.889113	0.077083	432.409094	0.000000
US_to_DE_no_retrain	robust_fixed_interval	0.948611	502.889113	0.179167	304.585743	-0.017143
US_to_DE_no_retrain	scenario_mpc	0.996528	6731.056651	0.000000	0.000000	-0.006749
US_to_DE_no_retrain	scenario_robust	0.982292	792.905592	0.000000	0.000000	-0.006749

6.8 Runtime and Intervention Behavior

DC^{3}S's computational overhead is modest: the reliability scoring, inflation calculation, and projection add <5% to the solver wall-clock time. The intervention rate (fraction of steps where the safe action differs from the optimizer's candidate) is 2.8% for DC^{3}S FTIT—meaning the optimizer's output is safe in >97% of steps, and the shield activates only when genuinely needed.

7 Governance and Reproducibility

The paper's quantitative claims are locked to versioned artifacts rather than free-text

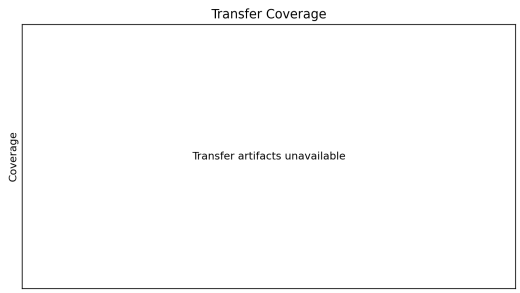

Figure 9: Transfer stress coverage summary across DE/US stress scenarios. DC^{3}S preserves safety more reliably than cost optimality under transfer/generalization stress.

manuscript statements. The locked DE and US impact summaries, the canonical stochastic run IDs, and the dashboard profile manifest are treated as publication interfaces. A claim is therefore valid only if it can be traced to a source artifact, a deterministic extraction rule, and a validator pass. This is the narrow sense in which governance is claimed as a contribution: reproducibility is embedded into the release procedure for this paper, consistent with the community push for stronger ML reproducibility norms [20].

Run-scoped evidence. The release process binds the manuscript to dataset-scoped report files, pinned run IDs, and a claim matrix. This prevents a common failure mode in long-running applied ML projects: a paper draft continues to circulate after the underlying evaluation artifacts have changed. In this draft, DE and US stochastic results are always reported with their run IDs, and region-level impact percentages are always read from the locked artifact family rather than recomputed manually.

What is and is not claimed. The governance contribution is operational, not philosophical. We do not claim a new general reproducibility framework for ML. We claim that for this particular CPS paper, metric locking, run provenance, and non-mutating consistency checks are necessary to keep the method story and the evidence story aligned.

Repository artifact traceability. Table 8 makes the repo-to-paper mapping explicit. It mirrors the paper structure used by this draft: the method section is implemented by train/eval code, experiments consume configs, seeds, and preprocessed data, results are derived from logs and checkpoints, and the paper figures and tables are generated by plotting and publication scripts.

Table 8: Repository artifact traceability for the paper pipeline.

Repo artifact family	Primary role in the paper	Main paper t
README + setup docs	Environment bootstrap, run commands, and operator setup	Governance pendix check
Train/eval code	Implements forecasting, uncertainty calibration, dispatch, and CPSBench evaluation	Method and
Configs + seeds	Fixes dataset scope, model settings, calibration settings, and randomized runs	Experimenta perparamete
Data + preprocessing	Defines dataset construction, split logic, and feature generation	Background/ mental setup
Logs + checkpoints	Preserve run IDs, solver outputs, metrics, and locked evidence values	Results nance/reproc
Plotting + publication scripts	Build figures, tables, and paper-facing assets from locked artifacts	Results and a

8 Discussion and Threats to Validity

8.1 Implications for CPS Safety

DC^{3}S demonstrates that *telemetry reliability should be a first-class input to the dispatch optimization loop*, not merely a pre-processing concern. The zero-violation result is achieved not by being maximally conservative (which would sacrifice cost), but by *adapting* conservatism to measurement quality. This principle generalizes beyond battery dispatch to any CPS where control decisions depend on degraded sensor inputs: HVAC systems, water treatment, autonomous vehicles under sensor failure.

8.2 Negative and Neutral Findings

We report several findings that qualify our claims:

1. **US peak shaving is 0.00%** in locked evidence—the US wholesale market structure in the evaluation window did not create peak-shaving opportunities.

2. **Solar high-regime PICP is 73.8%**, below the 90% nominal target. RAC-Cert compensates during dispatch, but better base calibration would reduce unnecessary intervention.

3. **Deep models do not currently outperform GBM** in the materialized rows of the promoted six-model comparison—GBM remains the strongest operational model across the published DE/US forecast surface.

4. **Cross-region impact magnitudes differ substantially**: a strategy yielding 7.11% cost savings in DE yields only 0.11% in US, underscoring regime sensitivity.

8.3 Threats to Validity and Limitations

1. **Dataset scope**: evidence is limited to DE (2018–2020) and US (2024–2026) windows. Generalization to other markets, tariff structures, or regulatory regimes requires additional evaluation.

2. **Simulation only**: all results are from software simulation with replayed telemetry. No hardware-in-the-loop or field deployment validation is claimed, and those hardware-facing stages remain future work only.

3. **Proxy assumptions**: cost and carbon signals are derived from publicly available data and may not fully represent all market settlement mechanisms.

4. **Stochastic sensitivity**: VSS and EVPI magnitudes are sensitive to scenario construction parameters and should not be interpreted as universal constants.

5. **Supporting proposition**: Proposition 1 provides marginal coverage preservation under inflation; it does not guarantee *conditional* coverage or optimal interval width.

9 Conclusion

We presented DC^{3}S, a conformal safety shield that integrates telemetry-reliability scoring, adaptive interval inflation (RAC-Cert), robust dispatch, action repair, and per-step certification into a single online loop for battery dispatch under degraded telemetry. Across four fault types and parametric severity sweeps on DE and US data:

- DC^{3}S achieves **zero true-SOC violations** (vs. 3.9% for deterministic and 25.6% for CVaR baselines) with only 2.8% intervention rate.
- Cost impact: 7.11% cost savings (DE), 0.11% (US); positive VSS in both regions.
- Ablation confirms RAC-Cert inflation and action repair are both necessary for zero-violation safety.

The practical message: trustworthy CPS dispatch requires telemetry-aware uncertainty calibration, not just robust optimization over assumed-correct uncertainty sets. DC^{3}S makes telemetry quality a *control input* rather than a hidden failure mode.

Future work.

1. Extend to hardware-in-the-loop validation with physical inverter/BMS integration as a future validation stage rather than as a claim of the current paper.
2. Improve base calibration on under-covered targets (solar high-regime) and make the main-text US analysis explicitly balancing-authority-specific across MISO, PJM, and ERCOT.
3. Add a clean deployment-facing latency/runtime benchmark and a publication-grade case trace showing telemetry degradation, interval widening, action repair, and safe dispatch over time.
4. Investigate conditional coverage guarantees under non-stationary degradation.

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A Hyperparameters and Configuration

Table 9: Core training and RAC-Cert hyperparameters.

group	key	value	source
train_forecast_de	horizon_hours	24	configs/train_forecast.yaml
train_forecast_de	lookback_hours	168	configs/train_forecast.yaml
train_forecast_de	cross_validation.n_folds	10	configs/train_forecast.yaml
train_forecast_de	cross_validation.gap	24	configs/train_forecast.yaml
train_forecast_us	horizon_hours	24	configs/train_forecast_eia930.yaml
train_forecast_us	lookback_hours	168	configs/train_forecast_eia930.yaml
train_forecast_us	cross_validation.n_folds	10	configs/train_forecast_eia930.yaml
train_forecast_us	cross_validation.gap	24	configs/train_forecast_eia930.yaml
dc3s_rac	sensitivity_probe	heuristic	configs/dc3s.yaml
dc3s_rac	sens_eps_mw	25.0	configs/dc3s.yaml
dc3s_rac	sens_norm_ref	0.5	configs/dc3s.yaml
dc3s_rac	beta_reliability	0.7	configs/dc3s.yaml
dc3s_rac	beta_sensitivity	0.5	configs/dc3s.yaml
dc3s_rac	k_sensitivity	0.4	configs/dc3s.yaml
regime_cqr	enabled	True	configs/uncertainty.yaml
regime_cqr	n_bins	3	configs/uncertainty.yaml
regime_cqr	vol_window	24	configs/uncertainty.yaml
regime_cqr	quantile_backend_policy	strict	configs/uncertainty.yaml

Key configuration values:

- Forecast: 24-hour horizon, 168-hour lookback, 10-fold CV with 24-hour gap.
- Conformal: $\alpha = 0.10$, volatility binning window = 168 hours.
- RAC-Cert: $\kappa_r = 0.5$, $\kappa_d = 0.3$, $\kappa_s = 0.2$, shrinkage $\tau = 0.95$.
- Drift detection: KS p-value threshold = 0.01, degradation threshold = 0.15.
- Optimization: cost weight $\lambda_c = 1.0$, carbon weight $\lambda_g = 1.2$, retraining cadence = 30 days.

B Additional Figures

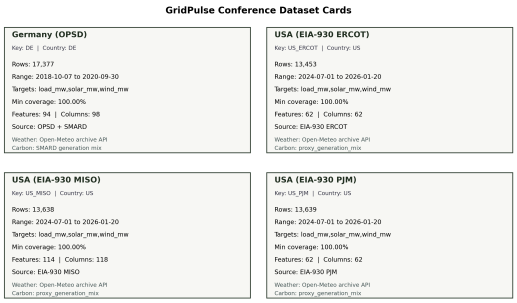

Figure 10: Dataset cards for all four regions in the release scope.

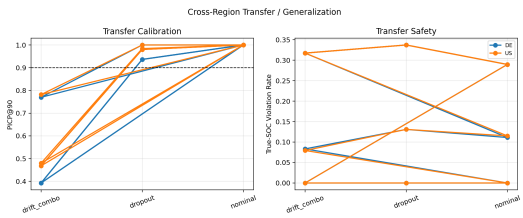

Figure 11: Cross-region transfer and generalization performance summary.

C Reproducibility Checklist

All results can be reproduced with the following commands:

```
# Train all models and build uncertainty artifacts
make train-all-ba

# Run CPSBench fault sweep
make r1-cpsbench

# Run ablation study
make r1-full

# Build publication artifacts and verify claims
make r1-verify
make paper-sync
```

Canonical run IDs: DE `20260217_165756`, US `20260217_182305`. All artifacts are hash-tracked in the release manifest.

D Operational Failure Modes

Table 10: Operational failure modes and safety controls.

Failure Mode	Trigger	Detection	Mitigation
Missing/stale data	Upstream delay	Readiness checks	Block u dispatch
Forecast drift	Distribution shift	KS/model monitors	Retraining workflow
Solver infeasible	Extreme intervals	Solver status flags	Safe infe response
Unsafe command	SOC/power violation	BMS validation	Reject wit tail
Control degradation	Heartbeat timeout	Watchdog detection	Lock re control